

PELATIHAN DAN PRA ASESMEN SERTIFIKASI CYBER SECURITY ANALYST

INSTRUKTUR
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CATRA INDO GROUP
Jakarta, 30 September – 2 Oktober 2020

IETF?

Internet Engineering Task Force (IETF)

• www.ietf.org

SLIDE 3

- Organisasi standar terbuka, yang mengembangkan dan mempromosikan standar Internet sukarela, khususnya standar yang terdiri dari paket protokol Internet (TCP/IP).
- Tidak memiliki daftar keanggotaan formal atau persyaratan keanggotaan. Semua peserta dan manajer adalah sukarelawan, meskipun pekerjaan mereka biasanya didanai oleh atasan atau sponsor.
- IETF dimulai sebagai kegiatan yang didukung oleh pemerintah federal AS, tetapi sejak 1993 telah beroperasi sebagai fungsi pengembangan standar di bawah naungan Internet Society, organisasi nirlaba berbasis keanggotaan internasional.

<https://www.ietf.org/rfc/rfc1035.txt>

- Tim Indeks Kerentanan Internet Domain Indonesia (KIDI) telah melakukan Penilaian terhadap konfigurasi dasar DNS Server pada domain **Telkom.co.id** sesuai dengan standar Internet Engineering Task Force (IETF) RFC 1035

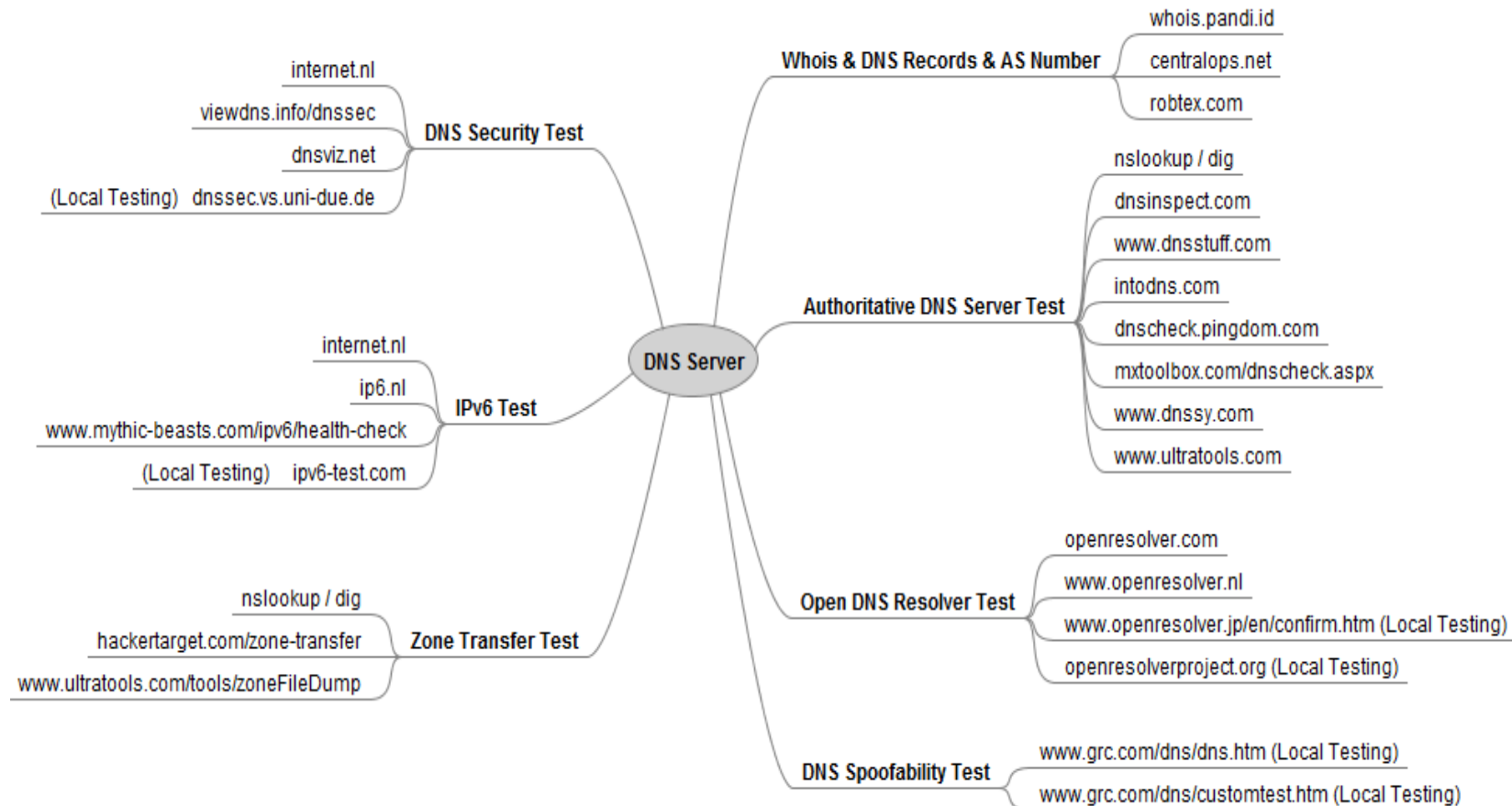
<https://www.ietf.org/rfc/rfc1035.txt>

Internet Engineering Task Force (IETF)

• www.ietf.org

SLIDE 5

- Bagan proses pengujian DNS Server dengan menggunakan INDONESE Assessment Framework v.0.93.2017

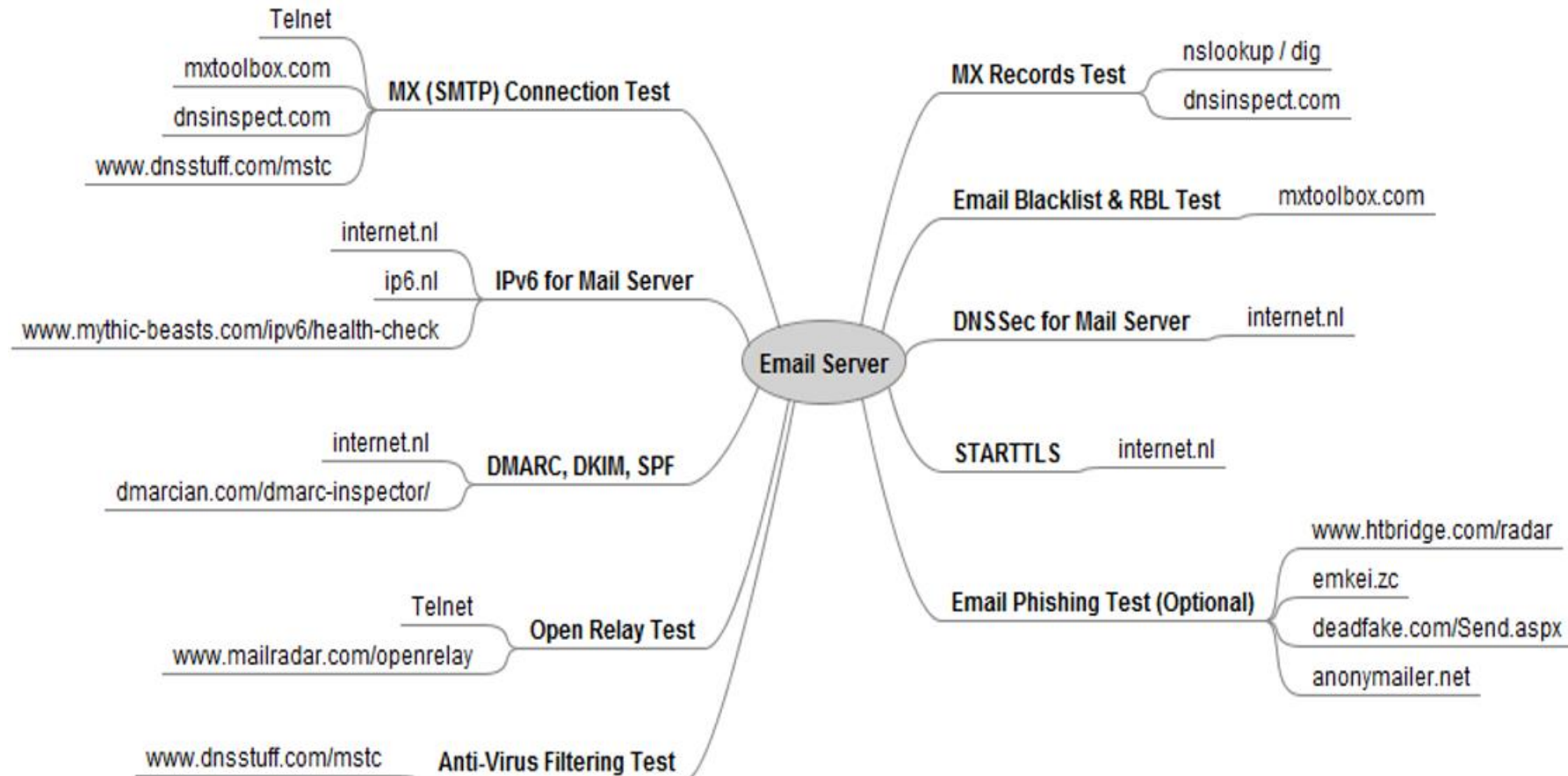


Internet Engineering Task Force (IETF)

• www.ietf.org

SLIDE 6

- Bagan proses pengujian Email Server dengan menggunakan INDONESE Assessment Framework v.0.93.2017

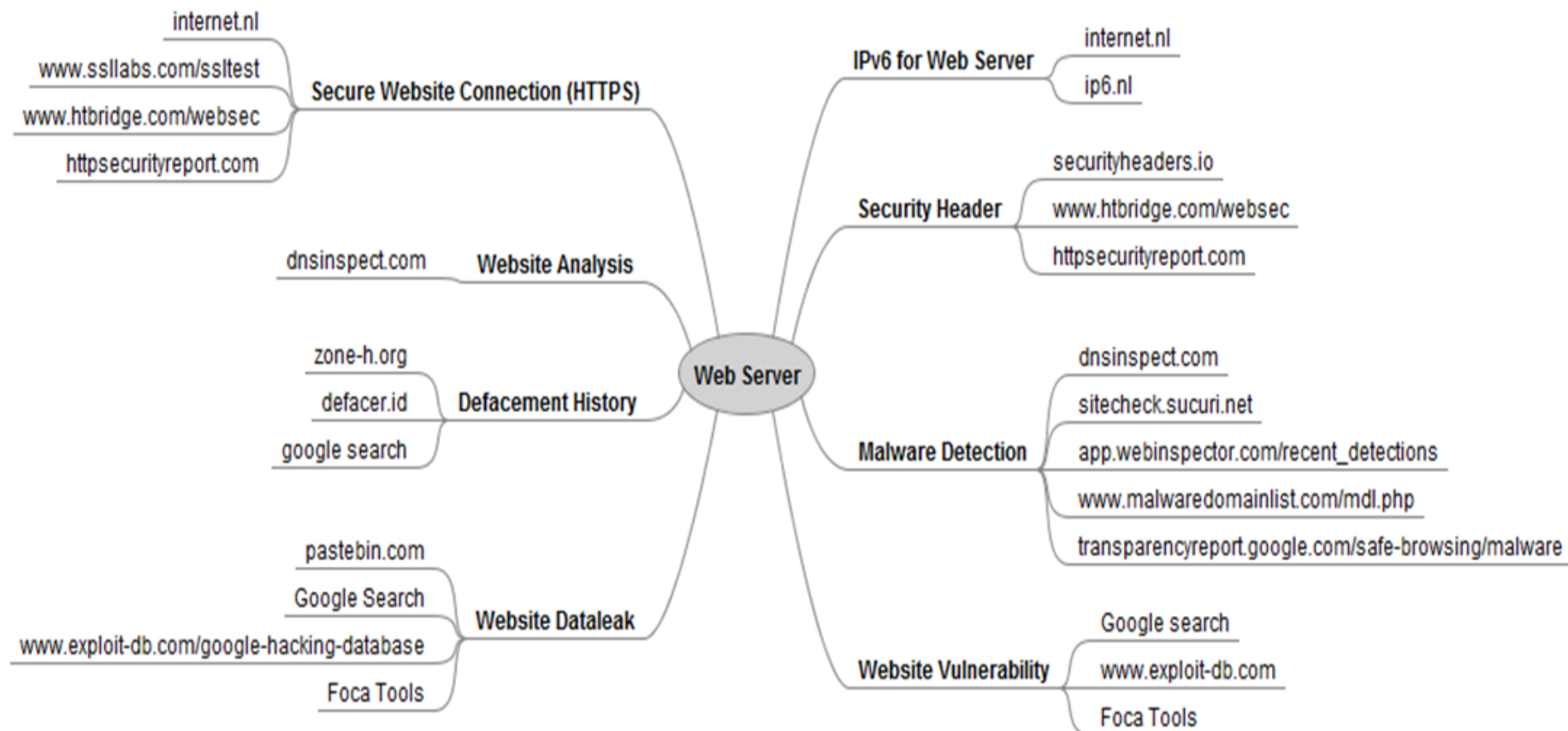


Internet Engineering Task Force (IETF)

• www.ietf.org

SLIDE 7

- Bagan proses pengujian Web Server dengan menggunakan INDONESE Assessment Framework v.0.93.2017



Proses Audit Domain dan DNS Server

SLIDE 8

Hasil Penilaian		
No	Kondisi eksisting Konfigurasi DNS Server	Hasil
1	Whois, DNS Record & AS Number Test	
2	Authorative DNS Server Test	
3	DNS Security Test	
4	IPv6 Test	
5	Open DNS Resolver Test	
6	Zone Transfer DNS Server Test	
7	DNS Spoofability Test (Optional) – Tidak dilakukan pengujian	

Keterangan :



: Konfigurasi sudah bersesuaian dengan best practice



: Terdapat ketidaksesuaian konfigurasi

B.1. Whois, DNS Record & AS Number

Tujuan:

Untuk mengetahui informasi umum tentang domain Telkom.co.id, antara lain :

- Tanggal dibuat, expire dan update domain terakhir,
- Registrant Contact/Admin Contact/Technical Contact/Billing Contact, Registrar.
- Name Server dan DNS Record
- Grafik Route Domain

Tools : Pandi Whois, <http://centralops.net/> dan <https://www.robtex.com/>

Proses Audit Domain dan DNS Server

SLIDE 10

Hasil Penilaian dan Rekomendasi		
No	Kondisi eksisting DNS dan Infrastruktur	Hasil
1	Data kontak Registrant, Admin, Technical dan Billing bersesuaian domain (Nama, alamat, telpon dan email).	✓
	Temuan 1 :	
	Rekomendasi :	
2	Terdapat 2 atau lebih Name Server yang bersesuaian dengan domain dan atau Memiliki Backup Name Server pada domain berbeda.	✓
	Temuan 1 :	
	Rekomendasi :	
3	Terdapat 2 atau lebih ASN dan Name Server berada pada subnet (ASNs) yang berbeda.	✓
	Temuan 1 :	
	Rekomendasi :	

Keterangan :



: Konfigurasi sudah bersesuaian dengan best practice



: Terdapat ketidaksesuaian konfigurasi

Proses Pengujian #1

- <http://whois.pandi.id/>

SLIDE 11

1. <http://whois.pandi.id/>

⚙ Data Domain

Domain ID:PANDI-D046576
Domain Name:TELKOM.CO.ID
Created On:18-May-1995 13:23:14 UTC
Last Updated On:13-Aug-2019 03:27:04 UTC
Expiration Date:01-Sep-2020 23:59:59 UTC
Status:ok
=====

Sponsoring Registrar Organization:Digital Registra
Sponsoring Registrar City:Sleman
Sponsoring Registrar Postal Code:55281
Sponsoring Registrar Country:ID
Sponsoring Registrar Phone:0274882257
Sponsoring Registrar Contact Email:info[at]digitalregistra.co.id
Name Server:NS3.TELKOMHOSTING.COM
Name Server:NS4.TELKOMHOSTING.COM
DNSSEC:Unsigned

who.is

WHOIS Search, Domain Name, Website, and IP Tools

telkom.co.id

Your IP address is 103.120.169.15

Looking to get a website? Web Hosting Website Builder SSL Certificates

Proses Pengujian #1

- <https://who.is>

SLIDE 12

who.is/whois/telkom.co.id

genda Pelatihan 2020 Beasiswa Bisnis Channel CV Elektro Catur Buku Erlangga Kuliah Komputer Penelitian Semi

who.is Search for domains or IP addresses... Premium Domains Transfer Features

telkom.co.id

whois information

Whois DNS Records Diagnostics

cache expires in 23 hours, 59 minutes and 57 seconds

Important Dates

Expires On	2021-09-01
Registered On	
Updated On	

Name Servers

ns3.telkomhosting.com	118.98.34.201
ns4.telkomhosting.com	203.130.232.85

Similar Domains

telko-api.com | telko-api.net | telko-elektronikka.com | telko-forum.com | telko-forum.de | telko-forum.net | telko-group.com | telko-gruppe.com | telko-gruppe.de | telko-kg.com | telko-lab.com | telko-lab.net | telko-maker.com | telko-maker.net | telko-medya.com | telko-net.biz | telko-net.com | telko-net.de | telko-net.info | telko-service.com |

Registrar Data

We will display stored WHOIS data for up to 30 days.

[Make Private Now](#)

Domain ID: PANDI-DO37911
Domain Name:telkom.co.id
Created On:1995-05-18 13:23:14
Last Updated On:2020-08-08 06:10:25
Expiration Date:2021-09-01 23:59:59
Status:ok
=====

Sponsoring Registrar PANDI ID:digitalreg
Sponsoring Registrar Organization:Digital Registra
Sponsoring Registrar City:Sleman
Sponsoring Registrar State/Province:Yogyakarta
Sponsoring Registrar Postal Code:55281
Sponsoring Registrar Country:ID
Sponsoring Registrar Phone:0274882257

Important Dates

Expires On	2021-09-01
Registered On	
Updated On	

Name Servers

ns3.telkomhosting.com	118.98.34.201
ns4.telkomhosting.com	203.130.232.85

Similar Domains

telko-api.com | telko-api.net | telko-elektronikka.com | telko-forum.com | telko-forum.de | telko-forum.net | telko-group.com | telko-gruppe.com | telko-gruppe.de | telko-kg.com | telko-lab.com | telko-lab.net | telko-maker.com | telko-maker.net | telko-medya.com | telko-net.biz | telko-net.com | telko-net.de | telko-net.info | telko-service.com |

Registrar Data

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Domain ID: PANDI-DO37911
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Created On:1995-05-18 13:23:14
Last Updated On:2020-08-08 06:10:25
Expiration Date:2021-09-01 23:59:59
Status:ok
=====

Sponsoring Registrar PANDI ID:digitalreg
Sponsoring Registrar Organization:Digital Registra
Sponsoring Registrar City:Sleman
Sponsoring Registrar State/Province:Yogyakarta
Sponsoring Registrar Postal Code:55281
Sponsoring Registrar Country:ID
Sponsoring Registrar Phone:0274882257
Sponsoring Registrar Contact Email:info@digitalregistra.co.id
Name Server:ns3.telkomhosting.com
Name Server:ns4.telkomhosting.com
DNSSEC:Unsigned

Information Updated: 2020-09-29 12:15:19

Proses Pengujian #1

centralops.net/co/

Apps Agenda Pelatihan 2020 Beasiswa Bisnis Channel CV Elektro Catur Buku Erlangga

Central Ops .net

Advanced online Internet utilities

Utilities

- Domain Dossier
- Domain Check
- Email Dossier
- Browser Mirror
- Ping
- Traceroute
- Nslookup
- AutoWhois
- AnalyzePath

Domain Dossier

Investigate domains and IP addresses

domain or IP address

☒ domain whois record ☒ DNS records ☐ traceroute

☒ network whois record ☐ service scan

user: anonymous [103.120.169.21]
balance: 49 units
[log in](#) | [account info](#)

Central Ops .net

Do you see Whois records that are missing contact information?
[Read about reduced Whois data due to the GDPR.](#)

Address lookup

canonical name **telkom.co.id.**

aliases

addresses **118.98.34.240**
180.250.116.128

Domain Whois record

Queried **whois.pandi.or.id** with "**telkom.co.id**"...

Domain ID: PANDI-DO37911
Domain Name: telkom.co.id
Created On: 1995-05-18 13:23:14
Last Updated On: 2020-08-08 06:10:25
Expiration Date: 2021-09-01 23:59:59
Status: ok

Network Whois record

Queried **whois.apnic.net** with "**118.98.34.240**"...

% Information related to '118.98.32.0 - 118.98.63.255'

% Abuse contact for '118.98.32.0 - 118.98.63.255' is 'abuse@telkom.co.id'

inetnum: 118.98.32.0 - 118.98.63.255
netname: TLKM_BB_INF_118_98
country: ID
descr: PT TELKOM INDONESIA

centralops.net/co/

Apps Agenda Pelatihan 2020 Beasiswa Bisnis Channel CV Elektro Catur Buku Erlangga

Central Ops .net

Advanced online Internet utilities

Utilities

- Domain Dossier
- Domain Check
- Email Dossier
- Browser Mirror
- Ping
- Traceroute
- Nslookup
- AutoWhois
- AnalyzePath

descr: PT TELKOM INDONESIA
descr: Menara Multimedia Lt. 7
descr: Jl. Kebonsirih No.12
descr: JAKARTA
admin-c: AR165-AP
tech-c: HM444-AP
remarks: -----
remarks: These IP was used for PT TELKOM Indonesia's Infrastructure.
remarks: -----
status: ASSIGNED NON-PORTABLE
mnt-by: MAINT-TELKOMNET
last-modified: 2009-02-20T12:33:51Z
source: APNIC

role: PT Telkom Indonesia APNIC Resources Management
address: PT. TELKOM INDONESIA
address: Menara Multimedia Lt. 7
address: Jl. Kebonsirih No.12
address: JAKARTA
country: ID
phone: +62-21-3860500
fax-no: +62-21-3861215
e-mail: ip-admin@telkom.net.id
admin-c: HM444-AP
tech-c: HM444-AP
nic-hdl: AR165-AP
notify: hostmaster@telkom.net.id
mnt-by: MAINT-TELKOMNET
last-modified: 2008-09-04T07:54:16Z
source: APNIC

person: PT Telkom Indonesia Hostmaster
nic-hdl: HM444-AP
e-mail: hostmaster@telkom.net.id
address: PT. TELKOM INDONESIA
address: Menara Multimedia Lt. 7
address: Jl. Kebonsirih No.12
address: JAKARTA
phone: +62-21-3860500
fax-no: +62-21-3861215
country: ID
notify: hostmaster@telkom.net.id
mnt-by: MAINT-TELKOMNET
last-modified: 2008-09-04T07:29:40Z
source: APNIC

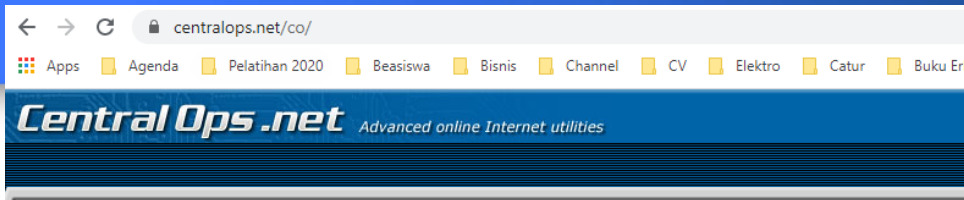
% Information related to '118.98.34.0/24AS17974'

route: 118.98.34.0/24
descr: PT. TELKOM INDONESIA
descr: JAKARTA
country: ID
origin: AS17974
mnt-by: MAINT-TELKOMNET
last-modified: 2015-05-27T03:33:03Z
source: APNIC

SLIDE 13

Proses Pengujian #1

SLIDE 14



Domain Dossier Investigate domains and IP addresses

domain or IP address

☒ domain whois record ☒ DNS records ☐ traceroute

☒ network whois record ☐ service scan

user: anonymous [103.120.169.15]
balance: 49 units
[log in](#) | [account info](#)

Do you see Whois records that are missing contact information?
[Read about reduced Whois data due to the GDPR.](#)

Address lookup

canonical name [telkom.co.id](#).

aliases

addresses **118.98.34.240**
180.250.116.128

Domain Whois record

Queried [whois.pandi.or.id](#) with "telkom.co.id"...

Domain ID: PANDI-DO37911
Domain Name:telkom.co.id
Created On:1995-05-18 13:23:14
Last Updated On:2020-08-08 06:10:25
Expiration Date:2021-09-01 23:59:59
Status:ok

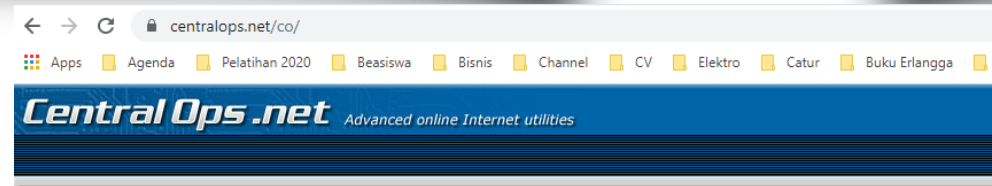
Network Whois record

Queried [whois.apnic.net](#) with "118.98.34.240"...

% Information related to '118.98.32.0 - 118.98.63.255'

% Abuse contact for '118.98.32.0 - 118.98.63.255' is 'abuse@telkom.co.id'

inetnum: 118.98.32.0 - 118.98.63.255
netname: TLKM_BB_INF_118_98
country: ID
descr: PT TELKOM INDONESIA
descr: Menara Multimedia Lt. 7
descr: Jl. Kebonsirih No.12
descr: JAKARTA



Domain Dossier Investigate domains and IP addresses

domain or IP address

☒ domain whois record ☒ DNS records ☐ traceroute

☒ network whois record ☐ service scan

user: anonymous [103.120.169.15]
balance: 49 units
[log in](#) | [account info](#)

Do you see Whois records that are missing contact information?
[Read about reduced Whois data due to the GDPR.](#)

Utilities

- Domain Dossier
- Domain Check
- Email Dossier
- Browser Mirror
- Ping
- Traceroute
- Nslookup
- AutoWhois
- AnalyzePath

admin-c: AR16S-AP
tech-c: HM444-AP
remarks: -----
remarks: These IP was used for PT TELKOM Indonesia's Infrastructure.
remarks: -----
status: ASSIGNED NON-PORTABLE
mnt-by: MAINT-TELKOMNET
last-modified: 2009-02-20T12:33:51Z
source: APNIC

role: PT Telkom Indonesia APNIC Resources Management
address: PT. TELKOM INDONESIA
address: Menara Multimedia Lt. 7
address: Jl. Kebonsirih No.12
address: JAKARTA
country: ID
phone: +62-21-3860500
fax-no: +62-21-3861215
e-mail: ip-admin@telkom.net.id
admin-c: HM444-AP
tech-c: HM444-AP
nic-hdl: AR16S-AP
notify: hostmaster@telkom.net.id
mnt-by: MAINT-TELKOMNET
last-modified: 2008-09-04T07:54:16Z
source: APNIC

person: PT Telkom Indonesia Hostmaster
nic-hdl: HM444-AP
e-mail: hostmaster@telkom.net.id
address: PT. TELKOM INDONESIA
address: Menara Multimedia Lt. 7
address: Jl. Kebonsirih No.12
address: JAKARTA
phone: +62-21-3860500
fax-no: +62-21-3861215
country: ID
notify: hostmaster@telkom.net.id
mnt-by: MAINT-TELKOMNET
last-modified: 2008-09-04T07:29:40Z
source: APNIC

% Information related to '118.98.34.0/24AS17974'

route: 118.98.34.0/24
descr: PT. TELKOM INDONESIA
descr: JAKARTA
country: ID
origin: AS17974
mnt-by: MAINT-TELKOMNET
last-modified: 2015-05-27T03:33:03Z
source: APNIC

% This query was served by the APNIC Whois Service version 1.88.15-SNAPSHOT (WHOIS-US3)

Proses Pengujian #2

SLIDE 15

2. <https://centralops.net>

name	class	type	data	time to live
telkom.co.id	IN	SOA	server: nspr1.telkom.net.id	86400s (1.00:00:00)
			email: hostmaster@telkom.net.id	
			serial: 2019083000	
			refresh: 3600	
			retry: 360	
			expire: 604800	
			minimum ttl: 86400	
telkom.co.id	IN	TXT	Telkom.Co.Id, PT Telekomunikasi Indonesia Official Corporate Domain	86400s (1.00:00:00)
telkom.co.id	IN	TXT	google-site-verification=CshEMU4RPE2TRT8NDO4C5Ty15bBf4RckZc4cH0RcDO0	86400s (1.00:00:00)
telkom.co.id	IN	TXT	MS=ms55499271	86400s (1.00:00:00)
telkom.co.id	IN	TXT	v=spf1 mx ip4:202.134.0.0/24 ip4:202.134.2.0/24 ip4:125.160.17.23 ip4:203.130.196.196 ip4:118.98.34.236 ip4:118.98.34.237 ip4:180.250.119.27 ip4:180.250.119.28 ip4:180.250.119.29 ip4:180.250.119.30 ip4:180.250.116.50 ip4:180.250.119.31 ip4:180.250.119.32 ip4:180.250.119.33 ip4:180.250.119.34 ip4:180.250.116.50 include:ncapp02.com ~all	86400s (1.00:00:00)
telkom.co.id	IN	MX	preference: 1	86400s (1.00:00:00)
			exchange: securemailgwjt3.telkom.co.id	
telkom.co.id	IN	MX	preference: 1	86400s (1.00:00:00)
			exchange: securemailgwjt4.telkom.co.id	
telkom.co.id	IN	NS	ns3.telkomhosting.com	86400s (1.00:00:00)
telkom.co.id	IN	NS	ns4.telkomhosting.com	86400s (1.00:00:00)

Proses Pengujian #2

SLIDE 16

centralops.net/co/

Apps Agenda Pelatihan 2020 Beasiswa Bisnis Channel CV Elektro Catur Buku Erlangga Kuliah Komputer Penelitian Seminar Internasio... Perguruan Tinggi Market Place Perusahaan Kemahasiswaan Other bookma

Central Ops .net

Advanced online Internet utilities

a service of Hexillio

Utilities About

Utilities

- Domain Dossier
- Domain Check
- Email Dossier
- Browser Mirror
- Ping
- Traceroute
- Nslookup
- AutoWhois
- AnalyzePath

descri: PT TELKOM INDONESIA
country: JAKARTA
ID
origin: AS17974
mnt-by: MAINT-TELKOMNET
last-modified: 2015-05-27T03:33:03Z
source: APNIC

% This query was served by the APNIC Whois Service version 1.88.15-SNAPSHOT (WHOIS-US3)

DNS records

DNS query for 240.34.98.118.in-addr.arpa failed: TimedOut

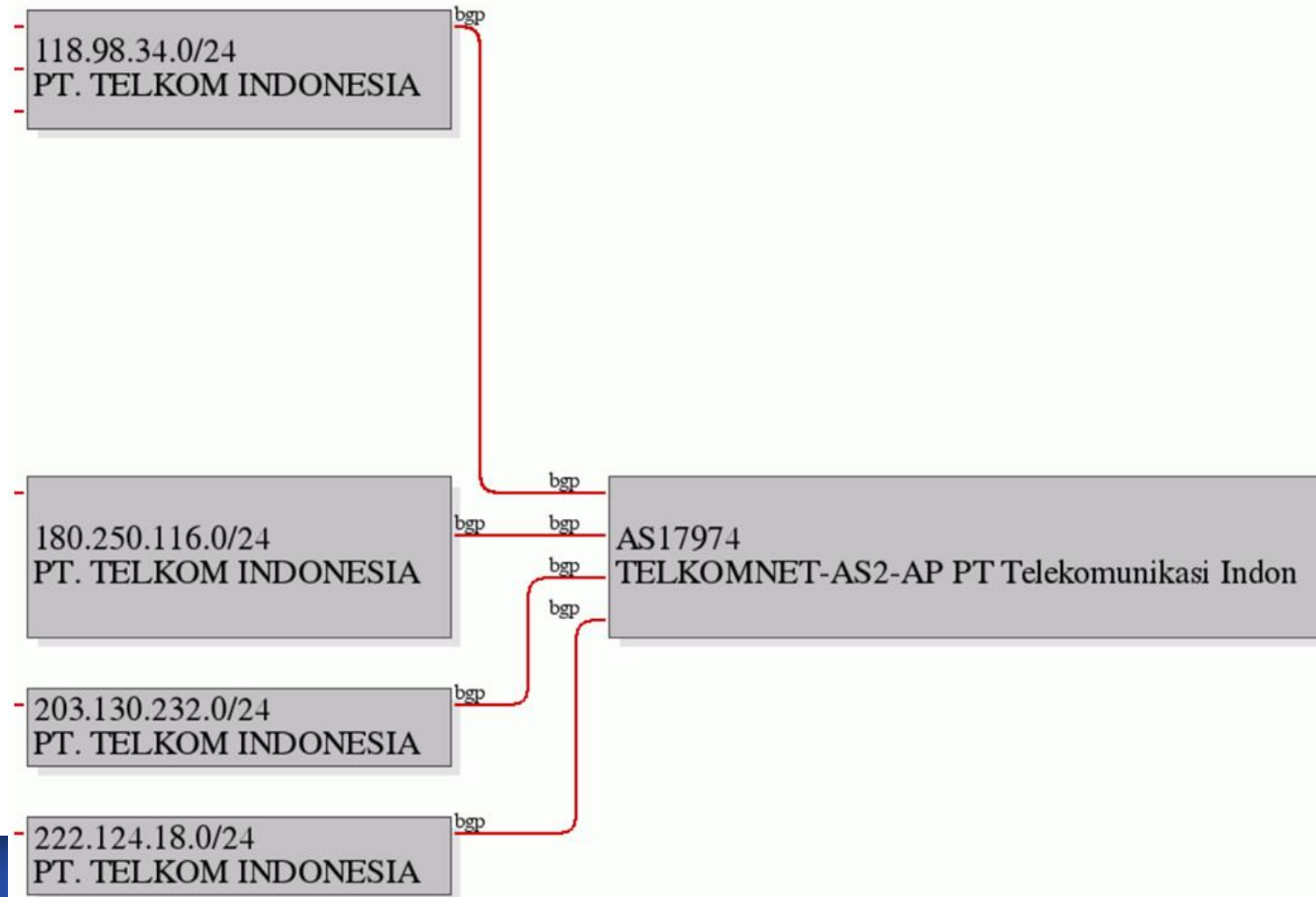
name	class	type	data	time to live
telkom.co.id	IN	A	180.250.116.128	3600s (01:00:00)
telkom.co.id	IN	A	118.98.34.240	3600s (01:00:00)
telkom.co.id	IN	NS	ns3.telkomhosting.com	86400s (1.00:00:00)
telkom.co.id	IN	NS	ns4.telkomhosting.com	86400s (1.00:00:00)
telkom.co.id	IN	MX	preference: 2 exchange: securemailgwjt4.telkom.co.id	3600s (01:00:00)
telkom.co.id	IN	MX	preference: 1 exchange: securemailgwjt3.telkom.co.id	3600s (01:00:00)
telkom.co.id	IN	TXT	MS=ms55499271	86400s (1.00:00:00)
telkom.co.id	IN	TXT	v=spf1 mx ip4:180.250.119.0/25 ip4:118.98.34.236 ip4:118.98.34.237 ip4:180.250.119.27 ip4:180.250.119.28 ip4:180.250.119.29 ip4:180.250.119.30 ip4:180.250.116.50 ip4:180.250.119.31 ip4:180.250.119.32 ip4:180.250.119.33 ip4:180.250.119.34 ip4:180.250.119.45 ip4:118.98.34.237 ip4:118.98.34.236 ip4:180.250.119.47 include:ncapp02.com -all	86400s (1.00:00:00)
telkom.co.id	IN	TXT	google-site-verification=CshEMU4RPE2TRT8NDO4C5Ty15bBf4RckZc4cH0RcDO0	86400s (1.00:00:00)
telkom.co.id	IN	TXT	Telkom.Co.Id, PT Telekomunikasi Indonesia Official Corporate Domain	86400s (1.00:00:00)
telkom.co.id	IN	SOA	server: nspr121.telkom.net.id email: hostmaster@telkom.net.id serial: 2020062244 refresh: 3600 retry: 360 expire: 604800 minimum ttl: 86400	86400s (1.00:00:00)

-- end --

Proses Pengujian #3

SLIDE 17

3. <http://robtex.com/>



Proses Pengujian #3

SLIDE 18

← → ↻ robtex.com/dns-lookup/telkom.co.id

Apps Agenda Pelatihan 2020 Beasiswa Bisnis Channel CV Elektro Catur Buku Erlangga Kuliah Komputer Penelitian Seminar Internasio... Perguruan Tinggi Market Place

telkom.co.id

telkom.co.id

ANALYSIS QUICK INFO REVERSE (NEW!) RECORDS SEO WOT ALEXA THREATMINER SHARED GRAPH HISTORY

ANALYSIS

This section shows a quick analysis of the given host name or ip number.

Telkom.co.id has two name servers, two mail servers and two IP numbers.

Telkomhosting name servers

The name servers are ns3.telkomhosting.com and ns4.telkomhosting.com.

Telkom mail servers

The mail servers are securemailgwjt3.telkom.co.id and securemailgwjt4.telkom.co.id.

IP numbers

The IP numbers are 118.98.34.240 and 180.250.116.128. The PTR of the IP numbers is 240.subnet118-98-34.astinet.telkom.net.id. The IP numbers are in Bekasi, Indonesia. They are hosted by PT. TELKOM INDONESIA.

Results found

Telkom.biz, **telkom.cc**, **telkom.center**, **telkom.ch**, **telkom.cloud**, **telkom.co**, **telkom.com**, **telkom.de**, **telkom.eu**, **telkom.fm**, **telkom.group** and **telkom.in**.

Proses Audit Domain dan DNS Server

SLIDE 19

Hasil Penilaian dan Rekomendasi		
No	Kondisi eksisting DNS dan Infrastruktur	Hasil
1	Data kontak Registrant, Admin, Technical dan Billing bersesuaian domain (Nama, alamat, telpon dan email).	✓
	Temuan 1 :	
	Rekomendasi :	
2	Terdapat 2 atau lebih Name Server yang bersesuaian dengan domain dan atau Memiliki Backup Name Server pada domain berbeda.	✓
	Temuan 1 :	
	Rekomendasi :	
3	Terdapat 2 atau lebih ASN dan Name Server berada pada subnet (ASNs) yang berbeda.	✓
	Temuan 1 :	
	Rekomendasi :	

B.2. Authoritative DNS Server Test

Tujuan :

Untuk mengetahui dan menilai sejauh mana penerapan konfigurasi DNS Server dan implementasinya pada domain Telkom.co.id sesuai dengan standar Internet Engineering Task Force (IETF) RFC 1035 <https://www.ietf.org/rfc/rfc1035.txt>

Tools : <http://www.dnsinspect.com/>

Hasil Penilaian dan Rekomendasi

No	Konfigurasi yang tidak bersesuaian	Hasil
1	Temuan 1 : Konfigurasi sudah sesuai	✓
	Rekomendasi :	
2	Temuan 2 :	
	Rekomendasi :	

1. <http://dnsinspect.com> dengan pilihan “Old DNS Inspect”

NS records [?]

NS records successfully fetched from the parent name server: d.dns.id

NS server	IPv4	IPv6	TTL
ns3.telkomhosting.com	222.124.18.184	?	43200
ns4.telkomhosting.com	203.130.232.85	?	43200

SOA record [?]

Primary name server [?]	nspr1.telkom.net.id
Hostmaster (e-mail) [?]	hostmaster.telkom.net.id
Serial [?]	2019083000
Refresh [?]	3600
Retry [?]	360
Expire [?]	604800
Minimum TTL [?]	86400

Proses Pengujian

- <http://dnsinspect.com>

SLIDE 22

dnsinspect.com/telkom.co.id/10446679

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Home Blog About Contact

Report for telkom.co.id

Input domain name ...

Run Report »

Report created on: Tue, 29 Sep 2020 12:01:48 GMT

Share this report: [Twitter](#) [Google+](#) [Facebook](#) [LinkedIn](#) | [Permissions](#)



✔ Parent	100
⚠ NS	97
⚠ SOA	94
⚠ MX	95
✖ Mail	60
✔ Web	100



PARENT

NS Records at Parent Servers

We have successfully fetched domain's NS records from parent name server ([e.dns.id](#)).
Domain NS records:

- ns3.telkomhosting.com. TTL=3600 [NO GLUE4] [NO GLUE6]
- ns4.telkomhosting.com. TTL=3600 [NO GLUE4] [NO GLUE6]

Missing Glue

Test ignored, name servers are located outside of current zone.

✔ Name Servers Have A Records

OK. Found A records for all name servers.

- [ns3.telkomhosting.com.](#) → 118.98.34.201
- [ns4.telkomhosting.com.](#) → 203.130.232.85

To reach your name servers via IPv4 an A record is needed for each name server.

🔔 Name Servers Have AAAA Records

NOTICE: While reading domain NS records at parent name servers, we found name servers without AAAA records.

- [ns3.telkomhosting.com.](#) → ?
- [ns4.telkomhosting.com.](#) → ?

To reach your name servers via IPv6 an AAAA record is needed for each name server.



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NS

NS Records

Your name servers returned 2 NS records:

- ns3.telkomhosting.com. TTL=86400 [118.98.34.201] [NO GLUE6]
- ns4.telkomhosting.com. TTL=86400 [203.130.232.85] [NO GLUE6]

✔ All Name Servers Responded

OK. All your name servers responded. We queried domain's records from all of your name servers and we received them successfully.

✔ Glue Check

OK. No differences found. The glue provided by the parent name servers has to match the data provided by the authoritative name servers.

✔ Allow Recursive Queries

OK. Domain name servers are not allowing recursive queries. On all name servers which acts as caching name servers recursive queries should be restricted to local networks. Having open DNS servers can lead to abuses such as cache poisoning and DOS (denial of service) attacks. Cache poisoning attacks allows under certain conditions to redirect legitimate web traffic, email and other traffic to malicious hosts compromising security.

✔ Check Name Servers Count

OK. Domain has 2 name servers. Recommended number, between 2 and 7 name servers (RFC 2182 recommends to have at least 3 authoritative name servers for domains).

✔ Identical NS Records

OK. All your name servers reported identical NS records. Each name server should return identical NS records.

✔ Check for Lame Name servers

OK. No lame name servers found. All of your name servers are configured to be either master or slave for your domain.

✔ Check All IPs are Public

OK. No private IPs found. Name servers using private IPs can't be reached from the Internet causing DNS delays.

✔ Name Servers Have A Records

OK. Found A records for each name servers.

- [ns3.telkomhosting.com.](#) → 118.98.34.201
- [ns4.telkomhosting.com.](#) → 203.130.232.85

To reach your name servers via IPv4 an A record is needed for each name server.

🔔 Name Servers Have AAAA Records

NOTICE: We found name servers without AAAA records.

- [ns3.telkomhosting.com.](#) → ?
- [ns4.telkomhosting.com.](#) → ?

B.3. DNS Security Test

Tujuan :

Untuk mengetahui apakah DNS Server yang digunakan sudah menerapkan DNS Security.

Tools : <https://internet.nl>, <http://dnsviz.net>, <https://dnssec-debugger.verisignlabs.com>

Hasil Penilaian dan Rekomendasi

No	Konfigurasi yang tidak bersesuaian	Hasil
1	DNSSEC signed	✗
2	DNSSEC validity	✗

Rekomendasi :

Keterangan :



: Konfigurasi sudah bersesuaian dengan best practice



: Terdapat ketidaksesuaian konfigurasi

Proses Pengujian

SLIDE 24

The screenshot shows the Internet.nl website interface. At the top, there's a navigation bar with links to various services like Apps, Agenda, Pelatihan 2020, Beasiswa, Bisnis, Channel, CV, Elektro, Catur, Buku Erlangga, Kuliah, Komputer, Penelitian, Seminar Internasional, Perguruan Tinggi, and Market Place. The main header features the Internet.nl logo with the tagline "IS YOUR INTERNET UP TO DATE?" and a language selector for English and Nederlands. Below the header, a prominent blue banner reads: "Modern Internet Standards provide for more reliability and further growth of the Internet. Are you using them?". The main content area is divided into three columns for testing: "Test your website", "Test your email", and "Test your connection". Each column includes a brief description of the test, a link to "about the test", a form for inputting domain or email addresses, and a "Start test" button. The bottom section contains three columns: "News" with recent updates, "Hall of Fame" listing domains with 100% scores, and "Statistics" providing counts for website, email, and connection tests.

internet.nl

Apps Agenda Pelatihan 2020 Beasiswa Bisnis Channel CV Elektro Catur Buku Erlangga Kuliah Komputer Penelitian Seminar Internasional Perguruan Tinggi Market Place

English Nederlands

Internet.nl
IS YOUR INTERNET UP TO DATE?

Home News Knowledge base Hall of Fame About Internet.nl

**Modern Internet Standards provide for more reliability and further growth of the Internet.
Are you using them?**

Test your website

Modern address? Signed domain? Secure connection? Security options?
[about the test](#)

Your website domain name:

Start test

Test your email

Modern address? Signed domain? Anti-phishing? Secure connection?
[about the test](#)

Your email address:

Start test

Test your connection

Modern addresses reachable?
Domain signatures validated?
[about the test](#)

Start test

News

- New version Internet.nl: X-XSS-Protection removed and improvement for no MX domains
- New TLS guidelines landed in Internet.nl
- Next major release of Internet.nl will use the new TLS guidelines
- Open source release Internet.nl including 'security headers'
- Email test on Internet.nl extended

Hall of Fame

779 domains with double 100%
Latest entry: 28-09-2020

- [digid.nl](#)
- [varoscak.de](#)
- [stedebroec.nl](#)
- [beatquantum.com](#)
- [hackthecat.com](#)
- [jwbrock.nl](#)
- [vanderbeekonline.nl](#)
- [computerbas.nl](#)
- [soverin.net](#)
- [vovert.nl](#)

Statistics

320838 website tests

- ✓ 100% score: 11232 websites
- ✗ 0-99% score: 309606 websites

121945 email tests

- ✓ 100% score: 2663 mail servers
- ✗ 0-99% score: 119282 mail servers

22988 connection tests

- ✓ 100% score: 6534 connections
- ✗ 0-99% score: 16454 connections

1. https://internet.nl/

✖ DNSSEC existence

Verdict:

Your domain is insecure, because it is *not* DNSSEC signed.

Test explanation:

We check if your domain is DNSSEC signed.

If a domain redirects to another domain via `CNAME`, then we also check if the CNAME domain is signed (which is conformant with the DNSSEC standard). If the CNAME domain is not signed, the result of this subtest will be negative.

Note: the validity of the signature is not part of this subtest, but part of the next subtest.

Technical details:

Domain	Registrar
telkom.co.id	PANDI Registry(for reserved names)

✖ Signed domain name (DNSSEC)

Too bad! Your domain is *not* signed with a valid signature ([DNSSEC](#)). Therefore visitors with enabled domain signature validation, are *not* protected against manipulated translation from your domain into rogue internet addresses. You should ask your name server operator (often your registrar and/or hosting provider) to enable DNSSEC.

[Show details](#)

✖ DNSSEC existence

Verdict:

Your domain is insecure, because it is *not* DNSSEC signed.

Technical details:

Domain	Registrar
telkom.co.id	None

Test explanation:

We check if your domain is DNSSEC signed.

If a domain redirects to another domain via `CNAME`, then we also check if the CNAME domain is signed (which is conformant with the DNSSEC standard). If the CNAME domain is not signed, the result of this subtest will be negative.

Note: the validity of the signature is not part of this subtest, but part of the next subtest.

🛡️ DNSSEC validity

Proses Pengujian

SLIDE 26

internet.nl/site/telkom.co.id/986203/

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Internet.nl
IS YOUR INTERNET UP TO DATE?

English Nederlands

Home News Knowledge base Hall of Fame About Internet.nl

Website test: telkom.co.id

32%

- ✗ Not reachable via modern internet address, or improvement possible (IPv6)
- ✗ Domain name not signed (DNSSEC)
- ✗ Connection not or insufficiently secured (HTTPS)
- ! One or more recommended application security options not set (Security options)

[Explanation of test report](#)

[Permalink test result \(2020-09-29 14:07 CEST\)](#)

[Seconds until retest option: 181](#)

Test another website

Your website domain name:

[Start test](#)

Directly test:

[www.telkom.co.id](#)

Test another email

Your email address:

[Start test](#)

Directly test:

[@telkom.co.id](#)

Test your connection

[Start test](#)

Further website testing

IPv6:

Modern address (IPv6)

Too bad! Your website is *not* reachable for visitors using a modern internet address ([IPv6](#)), or improvement is possible. Therefore your website is not part of the modern Internet yet. You should ask your hosting provider to fully enable IPv6.

[Show details](#)

Name servers

- ✗ IPv6 addresses for name servers
- IPv6 reachability of name servers

Web server

- ✗ IPv6 addresses for web server
- IPv6 reachability of web server
- Same website on IPv6 and IPv4

DNSSEC validity

Verdict:

This test did not run, because either a parent test that this test depends on gave a negative result ('fail') or not enough information was available to run this test.

Test explanation:

We check if your domain is signed with a valid signature making it 'secure'.

If a domain redirects to another signed domain via `CNAME`, then we also check if the signature of the CNAME domain is valid (which is conformant with the DNSSEC standard). If the signature of the CNAME domain is not valid, the result of this subtest will be negative.

Technical details:

Domain	Status
telkom.co.id	insecure

B.4. Ipv6 Test

Tujuan :

Untuk mengetahui apakah DNS Server yang digunakan pada domain Telkom.co.id sudah menerapkan Ipv6.

Tools : <https://internet.nl>

Hasil Penilaian dan Rekomendasi

No	Konfigurasi yang tidak bersesuaian	Hasil
1	IPv6 addresses for name servers	✗
2	IPv6 reachability of name servers	✗

Rekomendasi :

Keterangan :



: Konfigurasi sudah bersesuaian dengan best practice



: Terdapat ketidaksesuaian konfigurasi

1. https://internet.nl/

✖ IPv6 addresses for name servers

Verdict:

None of the name servers of your domain has an IPv6 address.

Test explanation:

We check if your domain name has at least two name servers with an IPv6 address. This is consistent with the [technical requirement of SIDN](#) (.nl TLD registry) that each .nl domain must have at least two name servers.

Technical details:

Name server	IPv6 address	IPv4 address
ns3.telkomhosting.com.	None	222.124.18.184
ns4.telkomhosting.com.	None	203.130.232.85

✖ Modern address (IPv6)

Too bad! Your website is *not* reachable for visitors using a modern internet address ([IPv6](#)), or improvement is possible. Therefore your website is not part of the modern Internet yet. You should ask your hosting provider to fully enable IPv6.

[Show details](#)

Name servers

✖ IPv6 addresses for name servers

Verdict:

None of the name servers of your domain has an IPv6 address.

Technical details:

Name server	IPv6 address	IPv4 address
ns3.telkomhosting.com.	None	118.98.34.201
ns4.telkomhosting.com.	None	203.130.232.85

Test explanation:

We check if your domain name has at least two name servers with an IPv6 address. This is consistent with the [technical requirement of SIDN](#) (.nl TLD registry) that each .nl domain must have at least two name servers.

IPv6 reachability of name servers

 IPv6 reachability of name servers

Verdict:

This test did not run, because either a parent test that this test depends on gave a negative result ('fail') or not enough information was available to run this test.

Test explanation:

We check if all name servers, that have an AAAA record with IPv6 adress, are reachable over IPv6.

Proses Audit Domain dan DNS Server

SLIDE 31

B.5. Open DNS Resolver Test

Tujuan :

Untuk mengetahui apakah DNS Server yang digunakan pada domain Telkom.co.id memiliki konfigurasi "Open DNS"

Referensi : <https://www.us-cert.gov/ncas/alerts/TA13-088A>

Tools: <http://openresolver.com/>

Hasil Penilaian dan Rekomendasi

No	Name Server Open DNS	Hasil
1	Tidak ada NS server dengan Open DNS	✓
2		

Rekomendasi :

Keterangan :



: Konfigurasi sudah bersesuaian dengan best practice

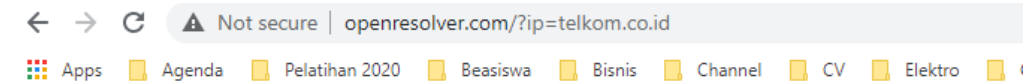


: Terdapat ketidaksesuaian konfigurasi

<http://openresolver.com/>

Recursive resolver is not detected on telkom.co.id

IP address telkom.co.id is **not vulnerable** to DNS Amplification attacks.



To manually test an IP address

```
dig +short test.openresolver.com TXT @1.2.3.4
```

(replace 1.2.3.4 with the IP address or domain name of the DNS server you are testing)

If you get "open-resolver-detected" in response, then you have a problem :)

Or, use a form:

Recursive resolver is not detected on telkom.co.id

IP address telkom.co.id is **not vulnerable** to DNS Amplification attacks.

Proses Audit Domain dan DNS Server

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B.6. Zone Transfer DNS Server (Domain Enumeration) Test

Tujuan :

Untuk mengetahui apakah zone transfer atau zone data atau zone file pada Authoritative DNS Server domain Telkom.co.id dapat ditampilkan.

Referensi :

https://en.wikipedia.org/wiki/DNS_zone_transfer

http://www.exploit-db.com/download_pdf/13687/

Tools : nslookup, dig, <https://hackertarget.com/zone-transfer/>

Hasil Penilaian dan Rekomendasi

No	Name Server yang rentan	Hasil
1.	Tidak ada NS dengan Zone Transfer enabled	✓
2.		

Rekomendasi :

Keterangan :



: Konfigurasi sudah bersesuaian dengan best practice



: Terdapat ketidaksesuaian konfigurasi

1. <https://hackertarget.com/zone-transfer/>

```
; <<>> DiG 9.11.3-lubuntu1.8-Ubuntu <<>> axfr @ns4.telkomhosting.com telkom.co.id
; (1 server found)
;; global options: +cmd
; Transfer failed.

; <<>> DiG 9.11.3-lubuntu1.8-Ubuntu <<>> axfr @ns3.telkomhosting.com telkom.co.id
; (1 server found)
;; global options: +cmd
; Transfer failed.
```

hackertarget.com/zone-transfer/

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HACKER TARGET SCANNERS TOOLS

Zone Transfer Online Test

Online Test of a **zone transfer** that will attempt to get all DNS records for a target domain. The zone transfer will be to

telkom.co.id

CHECK EXTERNAL ZONE TRANSFER

hackertarget.com/zone-transfer/

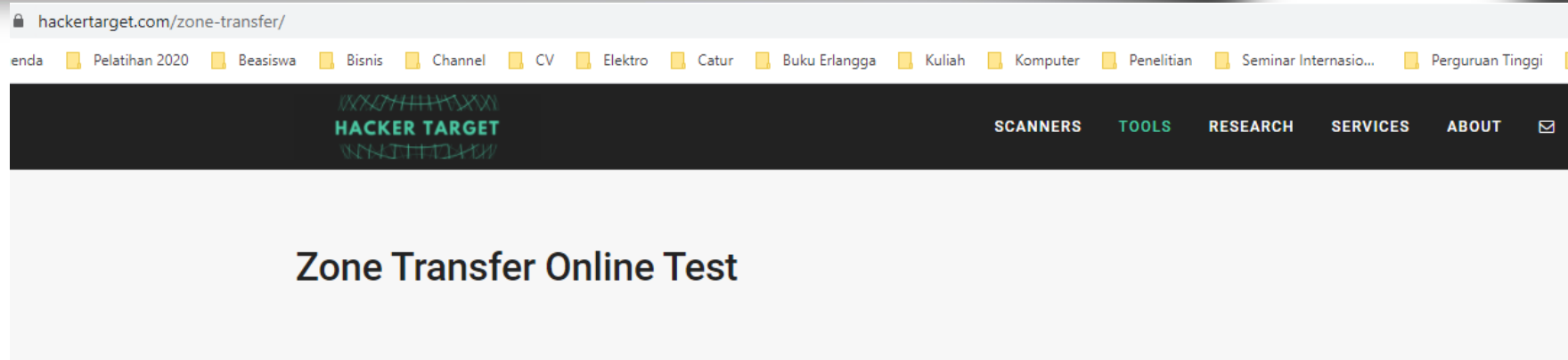
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HACKER TARGET SCANNERS TOOLS

Zone Transfer Online Test

Online Test of a **zone transfer** that will attempt to get all DNS records for a target domain. The zone transfer will be to

CHECK EXTERNAL ZONE TRANSFER



Online Test of a **zone transfer** that will attempt to get all DNS records for a target domain. The zone transfer will be tested against all name servers (NS) for a domain.

CHECK EXTERNAL ZONE TRANSFER

```
; <<>> DiG 9.11.3-1ubuntu1.13-Ubuntu <<>> axfr @ns3.telkomhosting.com telkom.co.id
; (1 server found)
;; global options: +cmd
; Transfer failed.
```

```
; <<>> DiG 9.11.3-1ubuntu1.13-Ubuntu <<>> axfr @ns4.telkomhosting.com telkom.co.id
; (1 server found)
;; global options: +cmd
; Transfer failed.
```

Berdasarkan beberapa pengetesan terhadap name server Telkom.co.id, level keamanan yang ada saat ini tergolong dalam level MEDIUM. Yang artinya, beberapa bagian dari Telkom.co.id sudah cukup bagus dari sisi security, namun ada beberapa bagian yang perlu diperbaiki untuk meningkatkan level keamanan domain Telkom.co.id.

Berdasarkan *finding* hasil analisa Name Server Telkom.co.id, berikut beberapa rekomendasi yang bisa dilakukan untuk meningkatkan tingkat Confidentiality, Integrity, Availability, dan security dari domain Telkom.co.id:

1. Implementasi IPv6 untuk domain Telkom.co.id
2. Implementasi DNSSEC untuk kewananan DNS